“Dump my brain so it remembers for me”: An AI-Augmented External Autobiographical Memory for Aphantasia and SDAM

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This is the accessible version of the printed poster. It contains all text on the poster, a description of the diagram, and the longer sentences that were shortened to fit the print.

- Printed poster (A0 PDF, visual layout)
- Paper: doi.org/10.1145/3797867.3841206 (resolves once the ACM proceedings are published)

Key message

> **I supply intent. The agent owns the rest of the memory loop: it gathers, connects, indexes, and surfaces, because I will not pull.**

My profile

Aphantasia: no voluntary mental imagery. **SDAM** (severely deficient autobiographical memory): I cannot relive my own past. Both are self-identified, not diagnosed.

Survey of Autobiographical Memory scores, out of 5

Domain	Score
Episodic	1.25
Future	1.17
Spatial	4.17

VVIQ-16: 21 of 80 (aphantasia range). One faculty is high, the rest are at the floor.

Why memory aids fail me

- **Nothing tells me what to record.** Faced with a blank diary, I stall.

- **I never return to read it.** The record, and the memory with it, withers.
- **Visual aids form no internal image**, and I rarely take photos.

The fix is not more storage but a system that elicits what to capture and returns it when it matters.

How the system works (diagram)

The poster shows a flow diagram with three boxes from left to right: Me (supply intent); AI agent, the largest box (sole reader, writer, organizer; gather, connect, index, surface); and Governed store (rules plus index). A Sources box at the top (messages, work, AI chats, calendar, health) feeds the store through an arrow labeled auto-ingest. Me sends intent to the agent and discusses with it; the agent pushes and surfaces context back to me. The agent records to and indexes the store, and reads the store first before acting. A dashed path from Me directly to the store, labeled “file or browse the store myself”, is marked as removed.

A loop I do not operate. The agent is the store’s only reader, writer, and organizer. The general-purpose agent (Claude) runs interactively and on a schedule, guided by per-folder rule files.

One month of use (2026-05-20 to 2026-06-20)

- **29 of 32** days with a captured daily entry
- **46** meeting notes (16 auto-pulled)
- **174** logged AI working sessions
- **27** items filed unconfirmed, **5** held back as too ambiguous to place

Descriptive counts from one participant, not efficacy measures. One auto-pulled meeting was misfiled, caught only at weekly review.

Four design principles from daily use

P1, the linchpin: Proactive recall and cross-source connection

Before a meeting, the agent assembled, unasked, the last outcome, open action items, and the messages in between: context from four or five places.

P2: Language and number as the channel

Long, disfluent, mixed-language speech goes in; prose and numbers come out. I read progress off data, such as knee flexion over weeks.

P3: Closure by articulation

I recap aloud; the agent confirms or corrects. One recap caught three misreadings in action items that had looked complete.

P4: Resurfacing dispersed threads

A question raised, set aside, and reopened over weeks was rebuilt into what was settled, rejected, and still open.

Connection is where the deficit bites

For a typical worker, fragmented context is a tax paid in toggling; the reconnecting still happens in the head. Mine does not, so the prosthesis must supply it. One risk does not invert: a confidently wrong output I cannot check against my own memory.

Limits

- N = 1, designer as participant.
- Needs a high digital footprint; a four-day trip cut capture to a fifth.
- Consent wins over coverage: nothing is recorded at home.

Poster C08 at ASSETS 2026, the 28th International ACM SIGACCESS Conference on Computers and Accessibility. All content is drawn from the paper.